Decision-Stable Causal Attribution: Complete Technical Document

Novelty assessment, formal problem definition, every system component, evaluation design, and phased plan

Course: Advanced Data Visualization Techniques
Domain: Marketing analytics, causal inference
Status: Research hypothesis, pending literature review and experimental validation

0. Thesis in One Paragraph

Standard and even recent causal attribution methods return a number per channel. A marketer then moves budget on that number, but the number depends on an assumption about hidden customer intent that the data cannot check. This project reports the **decision** (a budget allocation), the **confounding strength at which that decision breaks** (a breakdown point), and, when the decision is fragile, **which controlled experiment would most improve it**. Where the observational data cannot separate a channel's effect from pre-existing intent, the system says so instead of recommending.

The interface is the **Causal Constellation**: each channel is a star whose size is its causal credit, whose halo is its sensitivity interval, and whose outline says whether the decision is stable, fragile or unidentifiable (Section 5, C10).

1. Novelty Assessment

1.1 Overall verdict

This direction is clearly stronger than the earlier plans. It moves the contribution from "another attribution model" to "when can an attribution result be trusted for a decision, and what should we do when it cannot." That is a real practitioner problem and a defensible research framing. It is still a **hypothesis**, not an established gap, and the three capabilities are not equally novel.

1.2 Capability by capability

Capability	Assessment	Closest prior art (verify all in the literature review)	Main risk	Recommendation
1. Decision-stability analysis	**Most concrete and most defensible.** Sensitivity analysis usually asks whether an *effect estimate* survives hidden confounding. Asking whether a *budget decision* survives it, and reporting a breakdown point for the decision, is a sharper question.	Marginal sensitivity model (Tan), Rosenbaum bounds, E-values (VanderWeele and Ding), confounding-robust policy learning (Kallus and Zhou), partial identification (Manski)	Results may be vacuous: wide uncertainty sets make every decision "uncertain"	Make this the core contribution. Define the decision breakdown point precisely and show it is informative, not always "uncertain"

Capability	Assessment	Closest prior art (verify all in the literature review)	Main risk	Recommendation
2. Decision-driven experiment selection	**Most novel-sounding and hardest.** Choosing the experiment by expected decision improvement, within a sensitivity-based uncertainty set, for multi-touch marketing is a specific combination.	Bayesian experimental design, expected value of sample information, causal active learning, experiment-grounded debiasing (Kallus, Puli, Shalit), experiment-calibrated marketing mix models	Depends on a prior over outcomes that is itself an assumption. A weak implementation is easy to dismiss as textbook VOI	Keep it, but frame as "applying and validating VOI in this setting", and run a prior-robustness check
3. Evidence-based decision validation (overlap and sensitivity flags)	**Not novel on its own.** Overlap diagnostics and sensitivity analysis are standard.	Positivity checks, effective sample size, propensity trimming	Claiming it as a contribution invites an easy rejection	Reframe as an enabling component that feeds capabilities 1 and 2, not a separate contribution

1.3 What existing causal attribution work does and does not cover

Several papers already address confounding in multi-touch attribution, including CausalMTA, DCRMTA, and CAMTA. From their abstracts, they aim to *remove* confounding bias, mostly by learning representations or adjusting for user preference, so that the attribution output is unbiased. As far as I could confirm from abstracts only, they do not quantify how robust the resulting budget decision is to *hidden* confounding, and they do not recommend experiments. That is the plausible gap, but the abstracts are not the papers. The full text must be read before any claim is made.

A useful motivating result to look up: large field experiments comparing observational advertising measurement against randomized results have found that observational methods often fail to match the experiment (Gordon et al., Facebook field experiments).

That supports the problem, not your solution.

1.4 Weak points to address head-on

1. **The sensitivity parameter is not identifiable.** Any statement of the form "stable for Gamma up to 1.8" depends on how plausible 1.8 is. You need a principled way to calibrate it, such as benchmarking against the confounding strength of observed covariates.
2. **Conservatism.** Worst-case bounds can be too wide to be useful. The evaluation must show non-vacuous results.
3. **Three capabilities read as a bundle.** Present them as one pipeline with one thesis: report the decision, report where it breaks, say what fixes it.
4. **"Regret" on real data is not measurable.** Regret and decision quality can only be computed against known truth, which means the simulator and the randomized-data bridge in Section 6.

1.5 Honest positioning

A realistic outcome is a methodology with a good simulation study, a semi-synthetic real-data check, and a clear set of limits. That is a strong course project and potentially workshop-paper material. It is not a proven new method until the literature review and experiments say so.

2. Refined Research Questions and Hypotheses

ID	Research question	Hypothesis
RQ1	Do standard attribution methods and no-sensitivity causal estimators lead to wrong budget decisions under hidden intent?	Regret rises sharply with confounding strength for last-click, Markov, Shapley, and Gamma = 1 causal estimates.
RQ2	Does the decision breakdown point reliably separate trustworthy decisions from fragile ones?	Decisions flagged stable have a much lower wrong-decision rate than decisions flagged fragile.
RQ3	Is the stability analysis informative rather than vacuous?	In a meaningful share of realistic scenarios, decisions are certified stable at plausible Gamma.
RQ4	Does decision-driven experiment selection beat simpler policies?	It achieves lower regret per experiment than random, widest-interval, and largest-spend selection.

ID	Research question	Hypothesis
RQ5	Does the overlap diagnostic detect channels that observational data cannot identify?	High precision and recall on channels built to violate positivity in the simulator.

3. Formal Problem Definition

3.1 Notation

- Customer journeys i = 1..N, discrete time steps t = 1..T (for example days).
- Channels c = 1..K. Exposure indicator A_it,c (customer i exposed to channel c at step t).
- Observed covariates and history H_it (past exposures, clicks, timing, demographics or proxies).
- Hidden purchase intent U_it, unobserved, driving both exposure and conversion.
- Outcome Y_i (conversion within the window).

3.2 Estimand

Incremental effect of channel c: **tau_c = E[Y under intervention exposing c] - E[Y under intervention withholding c]**, other channels left at their natural behavior. Return per unit cost: **ROI_c = tau_c / cost_c**, with a concave saturation curve g_c in the simulator so allocation has interior solutions.

3.3 Decision

Budget B is allocated as b = (b_1..b_K) with minimum and maximum spend per channel. Value is V(b; tau) = sum over c of tau_c * g_c(b_c / cost_c). The optimal decision for a given tau is d*(tau) = argmax over b of V(b; tau), solved as a convex program.

3.4 Sensitivity model

Use the **marginal sensitivity model** with parameter **Gamma >= 1**: the odds of exposure given (H, U) differ from the odds given H alone by a factor within [1/Gamma, Gamma]. Gamma = 1 means no hidden confounding. For each Gamma, the estimator yields a per-channel interval [tau_c-(Gamma), tau_c+(Gamma)], and the joint uncertainty set T(Gamma) is the box of these intervals. The box ignores dependence between channels, so it is conservative; this is stated as a limitation.

3.5 Decision stability and the breakdown point

- **Point decision:** d_hat = d*(tau_hat at Gamma = 1).

- **Worst-case regret at Gamma:** MR(d; Gamma) = max over tau in T(Gamma) of [V(d*(tau); tau) - V(d; tau)].
- **Decision breakdown point:** Gamma* = smallest Gamma at which d_hat stops being optimal for some tau in T(Gamma), or at which MR(d_hat; Gamma) exceeds a tolerance epsilon chosen by the user.
- **Stability label:** "stable" if the plausible Gamma range sits below Gamma*, "fragile" otherwise, "unidentifiable" if the overlap check fails.
- **Robust decision:** d_rob(Gamma) = argmin over d of MR(d; Gamma), the minimax-regret allocation, reported alongside d_hat.

3.6 Experiment value

Candidate experiment e = (channel c, population slice s, holdout fraction h, sample size n, cost kappa_e). An experiment returns an unbiased estimate of tau_c in slice s with variance sigma^2(e) computed from the baseline conversion rate and n. After the experiment, the uncertainty set for channel c is intersected with the experiment's confidence interval, and the implied confounding strength for that channel is re-estimated.

Net value of an experiment: VOI(e) = E_outcomes[MR_before - MR_after(e)] - kappa_e, where the expectation is over experiment outcomes drawn from a working distribution inside T(Gamma). The selected experiment is e* = argmax VOI(e). The working distribution is an assumption, so every result is checked under at least two alternatives (uniform on the box, truncated normal around the estimate).

4. System Architecture

```
[Simulator]  [Real data: Criteo (randomized + observational)]
      \            /
   Journey store (Parquet): journeys, exposures, covariates, outcomes
                |
   Baselines: last-click, linear, Markov, Shapley, naive logistic
                |
   Causal estimator: propensity models -> AIPW / weighted MSM
                |
   Overlap diagnostics: ESS, propensity tails, trimming flags
                |
   Sensitivity engine: tau interval per channel across Gamma grid
                |
   Decision layer: d_hat, d_rob, regret, breakdown point Gamma*
                |
   Experiment selector: candidate designs, VOI ranking, sequential updates
                |
   Evaluation harness (simulator truth)   ->   results JSON
                |
   Visual interface (D3.js): Causal Constellation (hero), stability map,
allocation, overlap, experiment value
```

5. Components in Detail

C1. Synthetic customer-journey simulator

Purpose: the only place where truth is known, so everything is validated here first. A lazy simulator makes the whole study meaningless, so this is the most important component.

Design:

- Latent intent U_it: person-level baseline plus time drift (autoregressive), so intent can rise as a customer approaches purchase.
- Observed covariates X: noisy proxies of U with tunable quality rho. Lower rho means more hidden confounding.
- Conversion process: logistic hazard with an intercept, a term in U, and channel effects with decay (adstock) and saturation.
- Channels with deliberately different roles:
  - Branded search: targets high U, small true effect.
  - Retargeting: targets high U, small true effect.
  - Affiliate: targets high U, near zero true effect (the null channel).

    - Display prospecting: broad, low-U targeting, modest true effect.
    - Social: mid targeting, moderate effect.
    - Email: targets existing customers, moderate effect.
    - Video: broad targeting, positive effect.
- Exposure policies: each channel's exposure probability is a function of observed history and U, with a strength parameter per channel.
- Costs per exposure differ by channel.
- **Ground truth via interventions:** re-run the same random seeds and the same U while forcing a channel on or off, giving exact tau_c and exact optimal decisions.
- **Simulated experiments:** randomize exposure within a slice with a holdout fraction and return the outcome.

Knobs: sample size N, base conversion rate, confounding strength per channel, proxy quality rho, positivity floor (overlap), effect heterogeneity, saturation shape.

Named scenario suite: Mild confounding, Retargeting trap, Poor overlap, Hidden-intent heavy, Null channel, Heterogeneous effects.

Acceptance tests: known toy cases recover known effects with a fully randomized policy; the naive estimator is visibly biased when confounding is on; simulated outcomes reproduce the intended base rates.

C2. Data layer and schemas

- **journeys:** journey_id, user_id, outcome, window_start, window_end
- **exposures:** journey_id, t, channel, clicked, cost
- **covariates:** journey_id, t, feature columns (observed only; U never leaves the simulator)
- **truth (simulator only):** per-channel tau, optimal allocation, true Gamma per channel
- **results.json:** per scenario and method, the estimates, intervals across the Gamma grid, decisions, breakdown points, flags, and experiment rankings

Storage in Parquet, configs in YAML, fixed seeds for every run.

C3. Baseline attribution methods

Last-click, first-click, linear, Markov chain removal effect, sampled Shapley, and a naive logistic regression with channel counts. These exist only for comparison. Also include a simple propensity-adjusted version of the naive model to represent "causal-ish without sensitivity". If time allows, include a simplified reimplementation of one published causal MTA method as a stronger baseline, and state clearly that it is simplified.

C4. Causal effect estimator

Level 1 (MVP, point exposure per channel): define exposure as "exposed to channel c in the lookback window", with confounders measured before the first exposure. Estimate propensity with gradient boosting (cross-fitted), then use an augmented inverse-propensity-weighted (AIPW) estimator for tau_c, with weight truncation at a percentile.

Level 2 (full, time-varying): a marginal structural model with stabilized weights across time steps, with propensity at each step conditional on history. Optional GRU propensity model over sequences as a stretch.

Outputs: tau_hat_c, influence-function standard errors, weights, propensity scores.

Use cross-fitting to avoid overfitting bias in the nuisance models.

C5. Overlap and identifiability diagnostics

- Propensity histograms per channel for exposed and unexposed groups.
- Share of units with propensity below epsilon or above 1 minus epsilon.
- Effective sample size of the weights and the share of total weight held by the top 1 percent of units.
- Covariate balance (standardized mean differences) before and after weighting.
- **Rule:** if ESS or tail share crosses a threshold, label the channel "unidentifiable from observational data" and exclude its point estimate from the decision claim. Thresholds are tuned on the simulator and then fixed before real-data runs.

C6. Sensitivity engine

- Compute per-channel intervals under the marginal sensitivity model across a Gamma grid (for example 1.0 to 3.0 in steps of 0.1), using published sharp bounds for IPW and AIPW estimators (to be selected and checked in the literature review).
- **Calibration of Gamma:** benchmark against observed covariates, that is, leave one important observed covariate out, measure how much it shifts the estimate, and express hidden-intent strength as a multiple of that. This gives users a reference scale instead of a bare number.
- **Validity check in the simulator:** when the true Gamma is at or below the assumed Gamma, the interval should contain the true tau at least at the nominal rate.

C7. Decision layer

- Budget optimizer solving the allocation problem (convex, solved with a convex optimization library).

- Worst-case regret computed over the box T(Gamma). With a box and a concave separable objective, the inner maximization is tractable per channel.
- Outputs: d_hat, d_rob(Gamma), MR at each Gamma, the breakdown point Gamma*, and the stability label.
- Rank stability: Kendall tau between rankings at Gamma = 1 and at each larger Gamma.

C8. Experiment selector

- **Candidate set:** each channel crossed with a few slices (for example high-intent versus low-intent segments, new versus returning customers) and a few sample sizes, each with a cost.
- **Outcome model:** simulated or analytic distribution of the experiment's estimate given a working distribution of tau inside T(Gamma).
- **VOI computation:** Monte Carlo over outcomes, update the intervals, recompute MR, take the expected reduction minus cost.
- **Update step:** intersect the channel's interval with the experiment's confidence interval and re-estimate the channel's implied Gamma.
- **Sequential mode:** run the top experiment, update, repeat for k rounds.
- **Policies compared:** random, widest interval, largest spend, highest ranking-uncertainty, and the VOI policy.
- **Robustness:** repeat with at least two different working distributions.

C9. Evaluation harness

Runs every scenario with many seeds, applies every method, stores metrics, and produces tables and curves. Must be one command that regenerates every number in the report.

C10. Visual interface: the Causal Constellation

The hero view is the **Causal Constellation**: channels are stars on a dark canvas, and every glow, ring and pulse encodes a model output. The interface exists to make Gamma and decision fragility tangible, so a viewer can watch a channel's credit change as the measurement method or the assumed hidden-intent strength changes. It is not claimed as the novelty; the contribution is the methodology in Section 1.

Constellation encoding

Visual element	Encodes	Source
Star	One channel (5 to 7 stars)	C1, C2

Visual element	Encodes	Source
Star core size	Credit under the selected method: last-click, Markov, Shapley, or the causal estimate	C3, C4
Translucent halo	Sensitivity interval of the causal credit at the current Gamma; grows as Gamma rises	C6
Dashed ghost ring	Credit under a comparison method, so an over-credited channel shows a ghost ring far larger than its core	C3, C4
Outline style	Solid = stable, dashed = fragile, hollow with a question mark = unidentifiable from observational data	C5, C7
Pulsing star	The channel targeted by the highest-value experiment	C8
Star position	Force layout from co-occurrence of channels in journeys (context only)	C2
Arcs	How often two channels appear together in converting journeys; thin, muted, labeled as association, not causation	C2
True-effect tick (simulator only)	Ground-truth incremental effect	C1

Design rule: nothing glows, moves or pulses unless it encodes a model output.

Linked views

View	Shows	Technique
V1. Causal Constellation (hero)	Channels as stars with causal credit, sensitivity halos, ghost rings for standard methods, and stability outlines	Force layout, arcs, animated transitions, glow filters
V2. Decision stability map	Channels ranked by incremental ROI with intervals widening as Gamma increases, rank flips marked, breakdown point as a vertical line	Interval bump chart, area bands, Gamma slider

View	Shows	Technique
V3. Allocation under uncertainty	Recommended and robust allocations across Gamma, with the regret bound as a band	Stacked area or ribbon chart with uncertainty band
V4. Credit comparison	Per channel: last-click, Shapley, causal estimate, and (simulator only) truth	Dumbbell chart
V5. Overlap and identifiability	Propensity distributions per channel with flagged tails and ESS	Overlapping density plots, flag badges
V6. Experiment value	Candidate experiments ranked by net VOI, with before-and-after regret	Lollipop chart, linked to V1
V7. Benchmark dashboard (simulator)	Regret and error against confounding strength for every method	Line charts with confidence bands, small multiples by scenario

Interaction

- **Method switch:** last-click, Markov, Shapley, causal estimate. Stars resize with smooth transitions.
- **Gamma slider:** halos widen, rankings flip, and stars change outline from solid to dashed as the breakdown point is crossed. Reference markers show covariate-benchmarked Gamma values.
- **Click a star:** filters V2 to V7 to that channel.
- **Scenario selector:** swaps the simulator scenario (Retargeting trap, Poor overlap, and so on).
- **Run this experiment (simulator mode):** applies the top-ranked experiment, collapses the chosen star's halo, and updates the decision and regret figure.
- **Truth toggle (simulator mode):** shows ground-truth ticks.
- **Journey replay (stretch):** a comet travels one journey from star to star. Each touch is dimmed or brightened by its estimated exposure propensity, so touches that were nearly certain given the customer's history read as low-information.

The five demo moments

1. **Method switch.** Last-click to causal: retargeting, branded search and the null channel shrink while prospecting channels grow.

2. **Truth gap.** Ghost rings against cores show how much credit each standard method over-assigns.
3. **Gamma slider.** Halos widen and the ranking flips at the breakdown point; stars turn dashed.
4. **Unidentifiable channel.** A hollow star with a question mark: the data cannot separate the channel's effect from customer intent.
5. **Experiment pulse.** The pulsing star is the experiment to run. Running it collapses the halo and the decision turns stable.

Caveats for honest design

- Arcs and star positions are correlational context. Keep them muted and labeled, and allow hiding them.
- Ground-truth ticks and regret exist only in simulator mode; real-data mode shows no truth markers.
- The constellation is the hero view, and the stability map V2 stays as its precise companion, because rank flips and the breakdown point are easier to read there.
- Build a plain, working constellation first, then add glow and animation last.

Implementation: D3.js force layout, SVG for stars and halos with filter-based glow (canvas fallback if the arcs get heavy), reading the JSON contract defined in C2.

C11. Real-data module

- **Randomized bridge (most valuable):** use a large randomized advertising dataset (the Criteo uplift dataset is one candidate; confirm its license and structure). Create a deliberately confounded observational subsample by selecting treated units as a function of covariates, treat the full randomized result as ground truth, and check estimation error, interval coverage, and the stability flags against it. This gives real-data evidence without a hidden-truth problem.
- **Observational application:** apply the pipeline to the Criteo attribution dataset, where channels are anonymized campaigns, and report overlap flags and stability results descriptively. No causal truth claim.
- **Placebo tests:** use negative-control outcomes or pseudo-channels where the true effect should be zero.

6. Evaluation Design

6.1 Metrics

- **Estimation:** RMSE and bias of tau_c, interval coverage.

- **Ranking:** Kendall tau against the true ROI ranking.
- **Decision quality:** allocation regret, V(d*(tau_true); tau_true) minus V(d; tau_true).
- **Stability validity:** among decisions labeled stable, the wrong-decision rate; among fragile, the wrong-decision rate; informativeness as the share labeled stable.
- **Overlap detection:** precision and recall against channels built to violate positivity.
- **Experiment value:** regret after k experiments for each selection policy.

6.2 Experiments

ID	Experiment	Answers
E1	Estimation error across confounding strength, all methods	RQ1
E2	Allocation regret across scenarios, all methods	RQ1
E3	Stability labels against actual wrong decisions, plus sensitivity-interval coverage	RQ2, RQ3
E4	Experiment selection policies, regret versus number of experiments, with prior-robustness check	RQ4
E5	Overlap diagnostic detection	RQ5
E6	Randomized bridge on real data	Real-data support for E1 and E3
E7	Ablations: remove sensitivity (Gamma = 1), remove overlap flag, remove truncation, alternative Gamma benchmarking	Which parts matter

6.3 Statistical protocol

At least 200 simulation replicates per scenario and setting, fixed seeds, mean with 95 percent bootstrap intervals, paired comparisons between methods on the same seeds. All thresholds tuned on a separate set of simulator seeds, then frozen. Report negative results too.

7. Phases and Schedule

7.1 Honest scope statement

The complete system, including the time-varying estimator, experiment selection, the real-data bridge, and a polished interface, is roughly **2 to 3 weeks** of careful work. Coding agents speed up boilerplate, but validation and debugging of the statistics do not shrink much. With **3 days**, a credible MVP is possible, defined below.

7.2 Full phase plan

Phase	Work	Output	Effort
P0. Literature review and spec	Search the gap, read closest papers, freeze definitions of Gamma*, regret, VOI	2-page related work, frozen spec	1 to 2 days
P1. Simulator	Build C1 and its acceptance tests	Validated simulator, scenario suite	2 days
P2. Baselines and Level 1 estimator	C3, C4 (point exposure), C5	Baseline results, bias demonstration	2 days
P3. Sensitivity and decision layer	C6, C7	Intervals across Gamma, Gamma*, stability labels	2 days
P4. Experiment selection	C8	VOI ranking, sequential policy	2 days
P5. Evaluation	C9, E1 to E5, E7	Tables and curves	2 days
P6. Real-data bridge	C11, E6	Real-data evidence	2 days
P7. Time-varying estimator (stretch)	Level 2 of C4	Sequential MSM results	2 to 3 days
P8. Visualization	C10	Causal Constellation interface with linked views	3 days, can overlap P5 to P6
P9. Writing and demo	Report, video, repository cleanup	Final deliverables	2 days

7.3 Three-day MVP cut

Day	Include	Defer
Day 1	Short literature scan (2 to 3 hours), simulator with 5 to 6 channels and 3 scenarios, baselines	Time-varying estimator
Day 2	Level 1 AIPW estimator, overlap flags, sensitivity intervals, decision layer with Gamma*	Real-data bridge
Day 3	E1 to E3 with fewer replicates, V1 constellation with the method switch and Gamma slider, plus V2 stability map, short report	Experiment selection (add a simplified version only if time remains), V6 and V7

Minimum claim at the MVP: standard methods mis-allocate under hidden intent in simulation, and the decision breakdown point separates reliable from fragile decisions. Everything else is a labeled extension.

8. Technology and Repository

Python: NumPy, pandas, SciPy, scikit-learn, LightGBM, statsmodels, a convex optimization library for allocation, PyTorch only for the optional GRU, pytest for tests, YAML configs, DuckDB or Parquet for storage.
Front end: D3.js with Observable Plot for quick charts, static site reading the results JSON.
Reproducibility: one command to rebuild all results from seeds; pinned dependencies.

```
project/
  configs/            scenario and method YAMLs
  sim/                simulator and interventions
  methods/            baselines, estimator, overlap, sensitivity
  decision/           optimizer, regret, breakdown point
  experiments/        candidates, VOI, policies
  eval/               harness, metrics, plots
  data/               raw and processed (gitignored)
  viz/                D3 front end and results JSON
  tests/              simulator and estimator unit tests
  report/             write-up and figures
```

9. Risks and Mitigations

Risk	Mitigation
Prior work already covers the contribution	Do P0 first; adjust the claim to whatever gap is actually left
Stability analysis is vacuous (always "uncertain")	Report informativeness explicitly; tune decision tolerance epsilon; compare Gamma ranges against covariate benchmarks
Gamma cannot be validated on real data	Use the covariate-benchmark scale and the randomized bridge; state the limit
Simulator is too convenient	Include scenarios where the method should fail, and report them
Experiment selection depends on an arbitrary prior	Run at least two working distributions and show sensitivity
Time-varying estimator is unstable	Keep it a stretch; Level 1 carries the MVP
Scope too large for the time	Follow the 3-day cut and label extensions clearly
Overclaiming causality	Follow the claim rules in Section 11
Constellation arcs are read as causal links	Mute and label arcs as association, allow hiding them, keep all causal encodings on the stars
Visual effects consume build time	Build a plain constellation first, add glow and animation last, and keep the stability map as the fallback hero view

10. Literature Review Checklist

All names below are from memory and must be located and read, not cited blindly.

- **Causal multi-touch attribution:** CausalMTA, DCRMTA, CAMTA, early causally motivated attribution work (Dalessandro et al.).
- **Sensitivity analysis:** marginal sensitivity model (Tan), Rosenbaum bounds, E-values, omitted-variable-bias benchmarking (Cinelli and Hazlett), sharp bounds for IPW and AIPW under the marginal sensitivity model, sensitivity for marginal structural models.
- **Decisions under hidden confounding:** confounding-robust policy learning and evaluation (Kallus and Zhou), partial identification (Manski).

- **Experiments and observational data:** experimental grounding (Kallus, Puli, Shalit), data fusion, geo experiments, ghost-ad and holdout designs, large field experiments comparing observational and randomized ad measurement (Gordon et al.).
- **Marketing mix modeling with experiment calibration:** open-source MMM tools and their calibration approaches.
- **Experiment selection:** Bayesian experimental design, expected value of sample information, causal active learning.

Searches to run: "sensitivity analysis budget allocation causal", "value of information experiment selection marketing", "robust decision under unmeasured confounding advertising", "experiment design uncertainty marketing mix model calibration".

Deliverable of this phase: a table of each close paper, what it covers, and what the project adds beyond it. If a paper already covers capabilities 1 and 2 together, re-scope immediately.

11. Claim Discipline

You may claim:

- In simulation with known truth, method X mis-allocates budget under hidden intent and the decision breakdown point separates reliable from fragile decisions (once shown).
- In the randomized bridge, the pipeline recovers the randomized result within stated error under constructed confounding (once shown).

You may not claim:

- That the system finds the "true effect" on observational real data.
- That the stability result is independent of the Gamma assumption.
- That the framework is novel before the literature review is complete.
- That experiment selection is optimal, only that it beat named alternatives under stated assumptions.

12. Deliverables and Definition of Done

- Simulator with documented assumptions and passing tests
- Full pipeline code with a single reproduction command
- Results for E1 to E5 and E7 (and E6 if the bridge is completed)
- Causal Constellation interface with at least V1 (constellation), V2 (stability map), and V3 (allocation under uncertainty)
- Report with related-work table, method, results, limitations, and claim statement

- Three-minute demo video as a backup

Done means: every number in the report regenerates from the repository, every claim maps to an experiment, and the limitations section states what the data cannot show.